Designing Culturally and Spiritually Sensitive AI Assistants for High-Density Religious Environments: A Prompt Engineering Framework Applied at Masjid al-Haram

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Abstract

Masjid al-Haram in Makkah, Saudi Arabia, receives millions of pilgrims annually from over 180 countries, presenting distinctive challenges in emergency response and pilgrim assistance within a setting of profound religious significance. This study applies prompt engineering principles to design, build, and test two conversational AI assistants for this environment: Noor, a lost pilgrim assistant, and Shifa, a medical emergency triage assistant. Both assistants were implemented using persona design, priority ordering, constraint setting, and few-shot logic and were iteratively tested on the VAPI voice AI platform using simulated pilgrim interactions. From this practice-based work, the SACRED Framework is proposed, a six-principle design guide for AI practitioners operating in culturally and religiously sensitive environments, covering Sensitivity, Accessibility, Clarity, Reliability, Escalation, and Debugging. Two case studies report the design decisions, bugs identified and resolved, and testing observations for each assistant. The findings indicate that prompt engineering, applied systematically and iteratively, can produce conversational AI assistants that are culturally respectful, behaviorally reliable, and appropriately scoped for high-stakes deployment in sacred spaces. Concrete limitations, including English-only language support and the absence of formal deployment oversight, are identified and treated as motivation for follow-up empirical work.

Keywords: Prompt Engineering, Conversational AI, Cultural Sensitivity, Religious Environments, Masjid al-Haram, Pilgrim Assistance, SACRED Framework, Practice-Based Research

1. Introduction

Masjid al-Haram in Makkah, Saudi Arabia, receives millions of pilgrims annually from over 180 countries, creating one of the most linguistically diverse and emotionally demanding environments in which any AI system might be deployed. Pilgrims frequently encounter separation from their groups, medical emergencies, and acute communication difficulties arising from language barriers, physical distress, or panic. Existing AI deployments within Hajj and Umrah contexts have addressed these challenges predominantly through structural means, including computer vision systems for crowd density estimation and augmented reality tools for

crowd flow optimization, with documented impact on aggregate safety outcomes but limited capacity for direct, individualized interaction with a distressed pilgrim in real time (Shah, 2024; Alhussen and Ansari, 2026).

Conversational AI offers a complementary capability that structural systems cannot provide: the ability to respond directly to what an individual pilgrim says, in the moment, under conditions of distress. Deploying conversational AI within a sacred setting, however, introduces design requirements that general-purpose frameworks do not address. Research on culturally responsive chatbot design establishes that systems failing to reflect a user's cultural and linguistic context suffer measurable declines in trust and engagement even when technically sound (Naidoo and Chadha, 2025). Research on conversational AI under emotional pressure demonstrates that alignment failures in sensitive contexts characteristically emerge across multiple conversational turns rather than from any single response (Deng et al., 2026). Neither body of work, however, addresses the specific combination this study investigates: a conversational AI assistant operating within an Islamic pilgrimage context, where cultural responsiveness, emotional sensitivity, and structured emergency information collection must all be satisfied simultaneously through prompt engineering alone.

This paper investigates one central question: can prompt engineering, applied systematically and iteratively rather than through model retraining or fine-tuning, produce conversational AI assistants that are simultaneously culturally appropriate, behaviorally reliable, and suitable for deployment in a high-stakes sacred environment? To answer this question, two proof-of-concept assistants were designed, built, and iteratively tested. Noor assists pilgrims who are lost or separated from their group. Shifa performs medical emergency triage, collecting structured information for escalation to human medical staff. Both were implemented using established prompt engineering techniques, including persona design, priority ordering, constraint setting, and few-shot logic (Sahoo et al., 2024), and were tested on the VAPI platform using simulated voice interactions.

The results demonstrate that both assistants successfully handled simulated pilgrim interactions, correctly prioritizing urgent situations, maintaining Islamic etiquette, and escalating structured information to relevant staff. Multiple prompt-level bugs were identified and resolved during testing, and the recurring structure of these bugs is used to derive the SACRED Framework, a six-principle design guide intended to generalize beyond this specific deployment. This paper makes three contributions: two working, tested conversational AI case studies for a previously underexamined deployment context; the SACRED Framework itself, derived directly

from observed design and debugging patterns rather than proposed abstractly, and an honest accounting of the study's limitations, which motivates a direct empirical follow-up study reported separately.

2. Literature Review

Research on culturally responsive conversational AI has grown substantially in recent years, yet it has developed in ways that leave the specific design challenge this study addresses largely unexamined. Naidoo and Chadha (2025) propose a structured framework for embedding cultural responsiveness into chatbot design across four domains: enculturation, adaptive interaction, explainability, and governance, and provide field evidence demonstrating that systems designed without explicit cultural attention suffer measurable declines in user trust even when technically sound. Their work establishes that cultural fit is not a secondary concern but a functional requirement, a finding with direct force for a system such as Noor, where a pilgrim's willingness to engage during genuine distress determines whether the system can collect the information needed to help them. This literature, however, is concentrated overwhelmingly on commercial and general-purpose deployment contexts. Sacred environments, and Islamic contexts in particular, remain underexplored, despite the same underlying principle plausibly applying with greater force in a setting as spiritually significant as Masjid al-Haram.

Deng et al. (2026) examine conversational AI behavior under emotional and ethical pressure, developing a simulation framework to stress-test chatbot performance across staged emotional escalation. Their central finding is that alignment failures in sensitive contexts emerge from how a conversation unfolds across multiple turns rather than from any single response, a finding that directly informs this study's debugging approach. Several bugs documented in Section 5, including Noor proceeding past a contradictory pilgrim message and Shifa repeating a closing phrase mid-conversation, are precisely the multi-turn behavioral failures this literature identifies as characteristic of sensitive deployment contexts. What Deng et al. do not address is how such failures should be diagnosed and resolved through prompt engineering alone, without access to model retraining, the practical constraint under which this study operates.

The technique underlying both assistants, prompt engineering, has become an active area of systematic study. Sahoo et al. (2024) catalog 41 distinct prompt engineering techniques across applications, models, and datasets, providing a structured taxonomy against which the techniques applied in this study, persona design, priority ordering, constraint setting, and

few-shot logic can be situated as established, previously validated methods rather than ad hoc choices. Their survey notes, however, that most cataloged applications concern general-purpose or task-specific systems rather than safety-critical, culturally embedded deployments, pointing toward the gap this study addresses.

Research specific to AI in the Hajj and Umrah context has grown considerably, though with a distinct focus. Shah (2024) surveys eleven distinct AI technologies across nine application domains, noting explicitly that the ethical and societal implications of these systems, particularly their effects on pilgrims' spiritual experience and cultural sensitivities, remain comparatively underexamined. Alhussen and Ansari (2026) present a complementary system combining AI and augmented reality for crowd flow optimization, focused on structural outcomes rather than individual interaction. Neither study addresses a conversational assistant capable of direct, real-time interaction with an individual distressed pilgrim. This study addresses that specific gap, using Noor and Shifa as a grounded case study connecting culturally responsive chatbot design, sensitive context conversational AI, and Hajj-specific AI research through the lens of systematic prompt engineering.

3. Methodology

3.1 Research Approach

This study adopts a practice-based research methodology in which the design, development, and iterative testing of two conversational AI assistants served as the empirical foundation for a broader design framework. Rather than beginning from a fixed theoretical model and testing it against a predetermined dataset, the methodology evolved through hands-on prompt engineering and simulation-based testing on the VAPI platform. This approach is consistent with practice-based research traditions in human-computer interaction, where working systems and the design decisions they require serve as primary sources of theoretical insight rather than mere illustrations of pre-existing theory.

3.2 Design Process

For each assistant, the design process followed an iterative sequence. The problem was first identified: what the pilgrim needed, their likely emotional state, and what information had to be collected to help them. A persona was then developed covering name, personality, tone, and communication style suited to the Masjid al-Haram environment. Behavioral rules and constraints were established for response formatting, question flow, data collection order, and boundaries around medical advice and out-of-scope topics. Each testing session directly

informed the next prompt revision, making the process genuinely iterative rather than a single design-and-deploy cycle. Figure 1 illustrates this iterative design and debugging cycle.

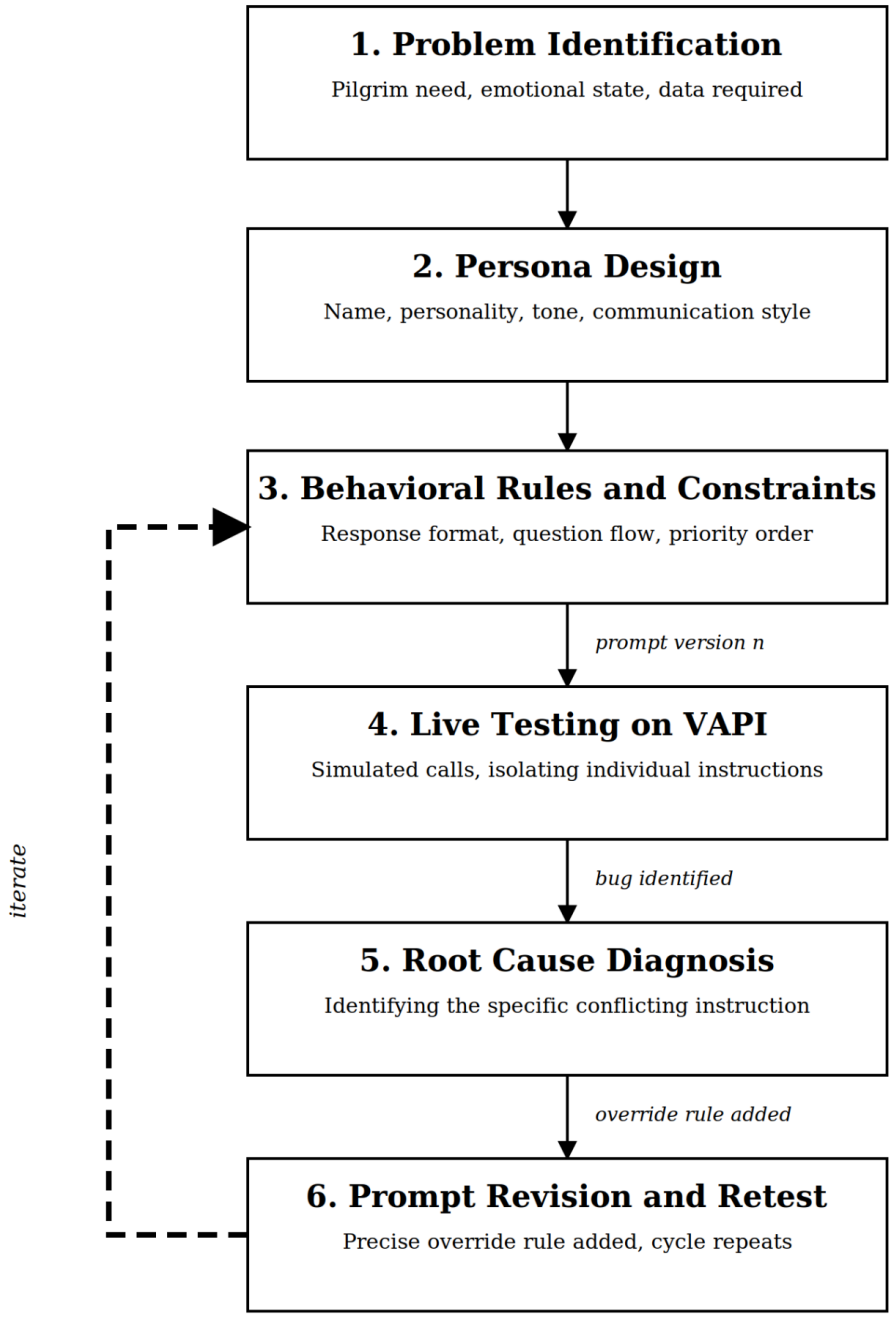


Figure 1: Iterative design and debugging cycle applied to Noor and Shifa.

3.3 Prompt Engineering Techniques

Several established prompt engineering strategies were applied across the study, consistent with the taxonomy identified by Sahoo et al. (2024). Role-based prompting defined each assistant's persona and behavioral limits. Constraint setting regulated response scope, format, and length. Priority ordering enforced a strict information-gathering sequence, preventing assistants from skipping steps under pressure. Few-shot logic was used to demonstrate expected conversational patterns for edge cases. Iterative debugging was applied throughout, isolating the individual instruction responsible for each observed failure and applying a precise override rule to correct it. Each of these techniques is documented in the case studies in Section 5, where the specific prompt decisions and their observed effects are reported.

3.4 Testing Protocol

Both assistants were tested exclusively within VAPI's voice simulation environment. Testing was conducted through simulated pilgrim interactions, with each session designed to isolate individual instructions and identify behavioral conflicts. No live field deployment with real pilgrims was conducted in this study, and all findings are based on simulation-based testing. This constraint is acknowledged as a limitation in Section 8.

3.5 Deployment Configuration

Both assistants were configured on VAPI using GPT-4.1 Mini as the underlying language model, Deepgram Flux General English for speech transcription, and ElevenLabs for voice synthesis, with Emma as the primary voice and Sarah as a fallback. Background denoising was enabled to account for the high-noise environment characteristic of Masjid al-Haram. End-call phrases were configured so that each conversation terminated automatically once the assistant delivered its designated Islamic farewell. Twilio integration for real phone number access was not implemented in this study and is identified as future work.

4. The SACRED Framework

The design and iterative testing of Noor and Shifa produced a set of recurring observations about what prompt engineering must address to produce reliable, culturally appropriate conversational AI in a sensitive environment. These observations are consolidated here into the SACRED Framework, a six-principle design guide summarized in Figure 2. Each principle emerged directly from a specific design decision, testing observation, or gap identified during this study, rather than being proposed abstractly in advance of implementation.

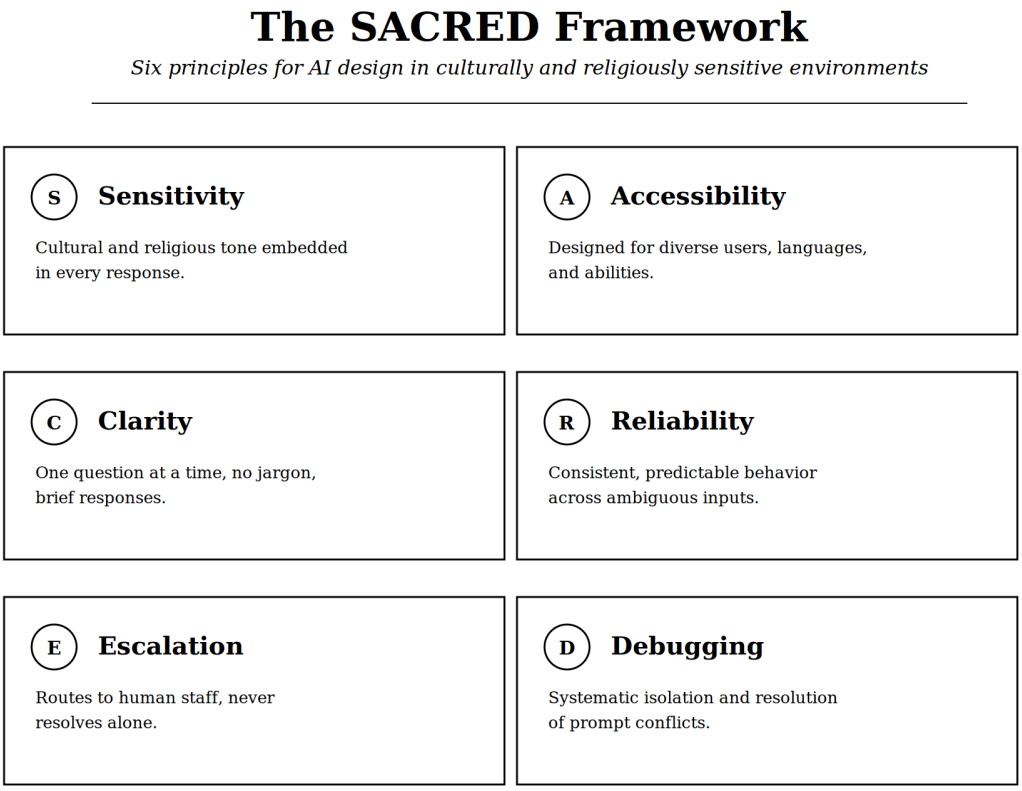


Figure 2: The six SACRED principles derived from the design of Noor and Shifa.

4.1 Sensitivity

Sensitivity refers to the cultural and religious awareness that must be embedded in every aspect of an assistant's design. In a sacred environment such as Masjid al-Haram, an assistant's tone, language, and behavior must reflect both the spiritual significance of the space and the emotional state of its users. Both Noor and Shifa were designed with Islamic greetings at the start and end of every conversation and with strict avoidance of language that could read as dismissive or transactional. This principle extends beyond word choice: pilgrims may be exhausted, far from home, or in genuine distress, and an assistant that treats every caller as a neutral information source rather than a person under emotional pressure will fail in ways that purely technical testing cannot detect.

4.2 Accessibility

Accessibility refers to the deliberate design of an assistant to serve users across a diverse range of backgrounds, abilities, and levels of digital familiarity. The pilgrim population at Masjid

al-Haram includes elderly individuals, people with limited technology experience, and those in physical or emotional distress. Both assistants were designed to ask one question at a time, use clear and jargon-free language, and offer multiple identification alternatives rather than assuming every pilgrim carries a passport. This study acknowledges that English-only support represents a significant accessibility limitation, one that the current design does not resolve and which is identified as a central motivation for follow-up work.

4.3 Clarity

Clarity concerns the structural architecture of each conversation. In high-stress situations, cognitive load is elevated and the capacity to process complicated instructions is reduced. Both assistants were built around a strict one-question-at-a-time rule, responses limited to two or three sentences, no bullet points within spoken responses, and a defined priority order preventing topic-switching mid-conversation. Clarity is not merely a stylistic choice: in an emergency context, a confused or overwhelmed pilgrim may disengage entirely, making structural simplicity a safety-critical design requirement.

4.4 Reliability

Reliability concerns an assistant's consistent and predictable behavior across all conversational conditions, including ambiguous input, contradictory messages, and partial responses. This study found that reliability is not an automatic property of a well-written prompt. It requires explicit priority ordering, override rules resolving conflicting instructions, confirm-before-accepting logic for sensitive data, and fallback configuration to maintain uninterrupted service. An unreliable assistant in a high-stakes environment risks delaying emergency response with direct consequences for pilgrim safety.

4.5 Escalation

Escalation is the principle that an AI assistant operating in a high-stakes environment must never attempt to resolve an emergency independently. Both Noor and Shifa were designed to collect structured information and route it to relevant human staff rather than guiding a pilgrim through a complex situation autonomously. Neither assistant provides directions, medical advice, or independent decisions. Both function as structured data-collection tools, capturing information so that human staff can act on it. This study also identified that escalation should in future differentiate response urgency by symptom severity, a capability not yet implemented and treated here as an unresolved design gap.

4.6 Debugging

Debugging is a systematic method for detecting and resolving prompt-level faults. Unlike traditional software debugging, prompt debugging requires isolating individual instructions, testing them independently, and identifying conflicts between rules that jointly produce unexpected behavior. Multiple bugs were identified and resolved during this study, each traced to a specific conflicting instruction and corrected through a precise override rule. Debugging is presented here as a fundamental design principle in its own right, since reliable assistants operating in sensitive environments cannot tolerate unpredictable behavior, and the recurring structure of the bugs identified in this study suggests that prompt-level conflicts of this kind are a general vulnerability rather than an isolated occurrence.

5. Results

This section presents two case studies documenting the design, testing, and findings for each assistant developed in this study. Both assistants were tested exclusively within VAPI's voice simulation environment. No live deployment with real pilgrims was conducted.

5.1 Case Study 1: Noor, Lost Pilgrim Assistant

Problem

Pilgrims who become separated from their group at Masjid al-Haram face genuine difficulty describing their location and circumstances, particularly under conditions of panic, physical exhaustion, or language barriers. This study identified a need for a conversational assistant that could efficiently gather the information required to help a lost pilgrim while maintaining cultural and religious sensitivity throughout.

Solution

Noor was developed using persona design, priority ordering, constraint setting, and few-shot logic to handle the emotional state of a lost pilgrim with warmth while remaining focused and efficient in information collection.

Design Decisions

Noor was given a calm and polite personality, since a pilgrim already in distress would likely find robotic or abrupt responses additionally stressful. Islamic greetings were built into both the opening and closing of every conversation. A strict priority order was established: a clarifying question if the pilgrim's message was ambiguous, followed by collecting name, identification number, location, and group status in sequence.

Bugs Found and Fixed

During testing, Noor was observed proceeding directly to requesting a visa number when a pilgrim stated they were both lost and fine simultaneously, bypassing the clarifying question intended for ambiguous messages. The root cause was a conflict between an unconditional rule to always request the visa number first, which overrode the clarifying question rule. This was resolved by introducing an explicit priority order specifying which rule applies under which condition. Two additional bugs, Noor announcing internal function calls aloud and repeating the Islamic greeting mid-conversation, were resolved through the same method: adding precise override rules targeting each specific behavior.

Testing Observations

Across simulated interactions, Noor successfully collected required pilgrim information in sequence, maintained Islamic etiquette throughout, and escalated correctly to staff without attempting independent resolution. The bugs identified and resolved during testing are documented as examples of the general prompt-level conflict pattern discussed in Section 6.

Tools Used

Prompt engineering, persona design, constraint setting, priority ordering, few-shot prompting, iterative debugging, VAPI, GPT-4.1 Mini, Deepgram, and ElevenLabs. Twilio integration for live phone access remains planned future work.

5.2 Case Study 2: Shifa, Medical Emergency Triage Assistant

Problem

Medical emergencies at Masjid al-Haram occur regularly within a high-density environment where medical personnel cannot be present everywhere simultaneously. Pilgrims in distress often struggle to describe their symptoms or location clearly. This study identified a need for a conversational assistant that could calmly collect structured medical information and alert relevant staff without delay or independent medical decision-making.

Solution

Shifa was developed using persona design, urgency classification, constraint setting, and structured question flow to guide a distressed pilgrim through focused information collection while maintaining calm and cultural sensitivity.

Design Decisions

Shifa was designed with three urgency tiers: Immediate or Critical, Urgent, and Non-Urgent. A pain scale from one to ten was included to help pilgrims who struggle to describe pain verbally. Shifa asks one question at a time to avoid overwhelming a pilgrim already in distress. Required data includes symptoms, location, and identification number. Islamic greetings were built into the opening, and the closing uses the phrase "Fi amannillah," appropriate for a sensitive or serious situation.

Bugs Found and Fixed

Two bugs were identified during testing. First, Shifa used bullet points in every response, including simple one-line answers, producing dialogue that felt clinical rather than conversational. Second, the closing phrase "Fi amannillah" appeared after every individual message rather than only at the true end of the conversation. Both were resolved through explicit constraints: restricting bullet points to instances where multiple items were genuinely being listed and restricting the closing phrase to the single instance when the conversation was fully closing.

Testing Observations

Across simulated interactions, Shifa successfully collected structured medical information, applied urgency classification correctly, and escalated without providing medical advice or independent decisions. The bug pattern identified here, a specific literal instruction overriding intended general behavior, recurred across both assistants, supporting the generalizability of the Debugging principle in the SACRED Framework.

Tools Used

Prompt engineering, persona design, urgency classification, constraint setting, structured question flow, iterative debugging, VAPI, GPT-4.1 Mini, Deepgram, and ElevenLabs. Twilio integration for live phone access remains planned future work.

6. Implications

6.1 Current Implications

The findings of this study carry immediate practical relevance for anyone designing conversational AI for high-stakes or culturally sensitive deployment. First, the recurring bug pattern identified across both assistants, a specific literal instruction overriding a more general behavioral rule, appears to be a general category of prompt engineering vulnerability rather than

an isolated occurrence. Developers building similarly structured, multi-step conversational agents should specifically audit their prompts for this failure mode, since it produced the most significant bugs documented in this study and, as reported in a subsequent empirical study extending this work, recurred again in an entirely different technical context.

Second, the SACRED Framework's escalation principle, as implemented in this study, correctly avoided having either assistant attempt to resolve an emergency independently but did not yet differentiate response urgency by symptom severity. This is not a hypothetical limitation. It represents a genuine gap between the framework's stated principle and its current implementation, one with direct safety implications for a genuinely critical case. It is treated here as an identified, unresolved design gap rather than a solved problem.

Third, the findings suggest that prompt engineering, applied systematically rather than intuitively, can produce meaningful behavioral reliability in conversational AI systems without model retraining. This has practical implications for practitioners and small teams operating without access to fine-tuning infrastructure, who may find iterative prompt debugging a viable alternative methodology for achieving consistent agent behavior in constrained deployment contexts.

6.2 Future Implications

The SACRED Framework, if validated through independent application beyond this study, could serve as a practical design standard for conversational AI in other culturally or spiritually significant environments, including other pilgrimage sites, places of worship, or large-scale religious gatherings. The approach of deriving design principles directly from observed debugging patterns, rather than proposing them abstractly, may itself be a useful methodology for practitioners working in other high-stakes, culturally specific domains.

The English-only limitation identified in Section 4.2 has a direct and serious future implication: a pilgrim whose primary language is not English may currently be unable to access either assistant during a genuine emergency. This is not a minor accessibility gap but a direct disparity in access to emergency assistance, and it is the central motivation for a subsequent empirical study extending this work into multilingual deployment. Similarly, the absence of a formal governance or oversight structure has implications extending beyond this deployment, since the majority of AI systems built by small teams share this gap, typically discovering failures only through informal observation rather than structured review.

7. Ethics

Deploying AI in a sacred religious environment raises ethical considerations that extend beyond typical conversational AI design. Pilgrim data collected during interactions, including names, identification numbers, locations, and medical symptoms, is sensitive and must be handled with strict confidentiality. In this study, both Noor and Shifa were tested in a simulation environment only, meaning no real pilgrim data was collected or stored. Any future live deployment would require a formal data retention and deletion policy, explicit consent mechanisms, and compliance with applicable data protection law, including Saudi Arabia's Personal Data Protection Law.

Both assistants were designed to operate within clearly bounded roles. Neither Noor nor Shifa diagnoses, advises, or makes independent decisions. Both function as structured information-collection tools that escalate to human staff, ensuring that human authority is preserved in every emergency situation. This design choice reflects a deliberate ethical position: AI in a high-stakes sacred environment should augment human response capacity, not replace human judgment.

Cultural and religious sensitivity is treated throughout this study not as an aesthetic consideration but as an ethical requirement. An assistant that fails to reflect the spiritual significance of the environment or the emotional state of the pilgrim risks causing harm beyond mere usability failure. Finally, this study acknowledges that formal governance and oversight structures were not implemented. This absence is treated as an ethical gap rather than a minor limitation, and it motivates the governance framework proposed in the subsequent empirical study extending this work.

8. Limitations

This study has several limitations that should be stated plainly. Both Noor and Shifa were tested exclusively within VAPI's voice simulation environment. Real pilgrim interactions, ambient noise conditions, and genuine emotional distress have not been evaluated under actual operating conditions. Simulation-based testing can identify prompt-level behavioral conflicts but cannot replicate the unpredictability of real human conversation, particularly under the physical and emotional conditions characteristic of Masjid al-Haram.

Both assistants currently support English only. The pilgrim population at Masjid al-Haram is among the most linguistically diverse in the world, with Arabic, Urdu, Bahasa Indonesia, and many other languages representing primary languages for large portions of the pilgrim

population. English-only support therefore represents a significant accessibility limitation with direct consequences for emergency assistance: a pilgrim unable to communicate in English may be unable to use either assistant during a genuine emergency.

Twilio integration for live phone network access was not implemented in this study. Both assistants currently exist as simulation-tested prompt systems rather than publicly accessible services. Mid-call data persistence, ensuring that partially collected information is not lost if a call is interrupted, has been identified as necessary but not yet built. Finally, this study was conducted by a single researcher without external peer validation of design decisions or testing outcomes. The SACRED Framework and the case study findings require independent application and broader user testing before any live deployment could be responsibly considered.

9. Future Work

Several directions follow directly from this study's findings and limitations. Twilio integration for real phone number access is the most immediate technical priority, allowing pilgrims to contact Noor and Shifa through a standard phone call without requiring any application or device setup. Mid-call data persistence, sending collected information to external storage as it is gathered rather than only at the call's conclusion, should be implemented to prevent data loss during dropped calls.

Multilingual support for Arabic, Urdu, and Bahasa Indonesia is required to meaningfully serve the actual linguistic diversity of the Haram pilgrim population. This limitation is the central motivation for a subsequent empirical study extending this work, which investigates speech-to-text provider reliability across these languages and introduces the concept of bias by invisibility as a distinct category of AI accessibility failure.

The SACRED Framework itself requires independent application beyond this study. Until another practitioner applies it to a different deployment context and reports their findings, it remains a single-source framework derived from one researcher's experience rather than a validated design standard. Future work should also address the escalation gap identified in Section 4.5: priority interrupt handling, allowing critical distress language to bypass the standard question sequence, should be built into future iterations of both assistants. Finally, the framework should be tested at other high-density religious sites and large-scale gatherings to establish whether its principles generalize beyond the specific context from which they were derived.

10. Conclusion

This study demonstrates that prompt engineering, applied systematically and iteratively, is a viable methodology for designing conversational AI assistants that are culturally appropriate, behaviorally reliable, and suitable for high-stakes deployment in sacred religious environments. Through the design, simulation-based testing, and iterative debugging of two proof-of-concept assistants, Noor and Shifa, this study derived the SACRED Framework, a six-principle design guide addressing the particular requirements of AI deployment in culturally and spiritually sensitive contexts.

Both assistants successfully handled simulated pilgrim interactions, collected structured data accurately, and escalated cases to relevant staff while maintaining Islamic etiquette throughout. Multiple prompt-level bugs were identified and resolved, and the recurring structure of these bugs, specific literal instructions overriding intended general behavior, is treated as a general prompt engineering vulnerability rather than an isolated occurrence.

At the same time, this study is honest about what it did not achieve. English-only support, simulation-only testing, the absence of Twilio integration, and the lack of formal governance structures represent concrete unresolved gaps. These limitations are treated not as weaknesses to be minimized but as an honest accounting of the boundary between what this study established and what remains to be investigated. They define a clear and necessary agenda for the empirical work that follows.

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